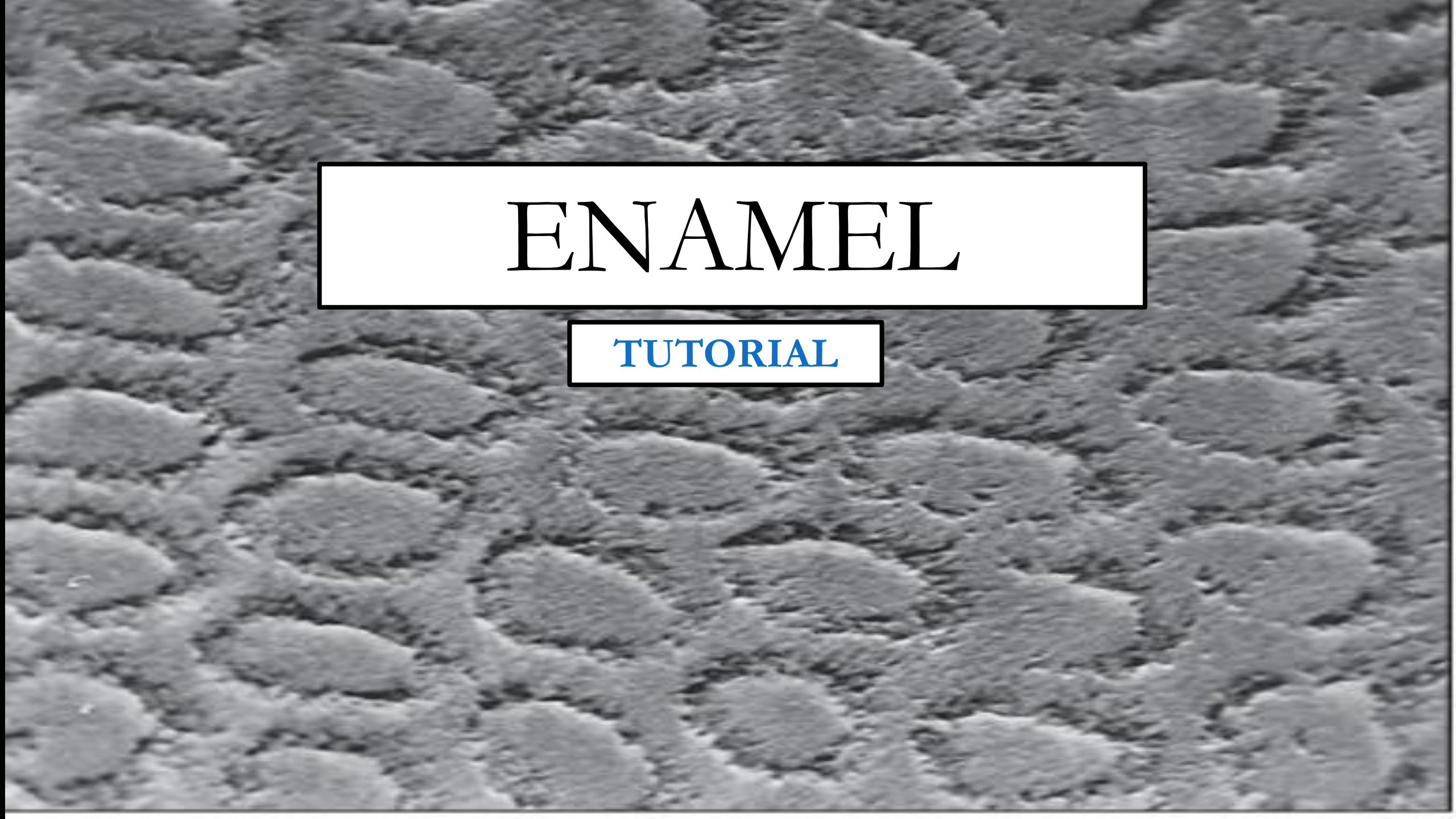


The background of the slide is a grayscale scanning electron micrograph (SEM) showing the intricate, textured surface of tooth enamel. The surface is composed of numerous small, rounded, and slightly irregular features, giving it a rough, crystalline appearance. The lighting creates highlights and shadows that emphasize the three-dimensional nature of the enamel's structure.

ENAMEL

TUTORIAL



TASK 1: STUDY, PASTE, AND LABEL THE FOLLOWING FIGURES

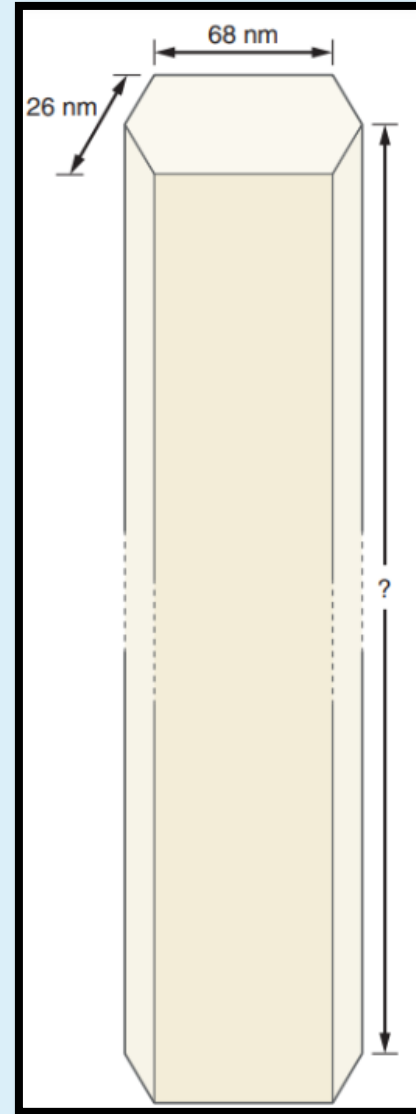
**Note: please write the figure heading and labelling
by hand in your log books. No need to write any
other descriptive points**

The dimensions and shape of an enamel crystal (hexagonal) (Fig 7.5)

Enamel crystals → *long flat ribbons rhomboidal* in cross section

Crystal core → ↑ *soluble than periphery* (richer in magnesium and carbonate)

Enamel crystallites → *regularly hexagonal* in cross section



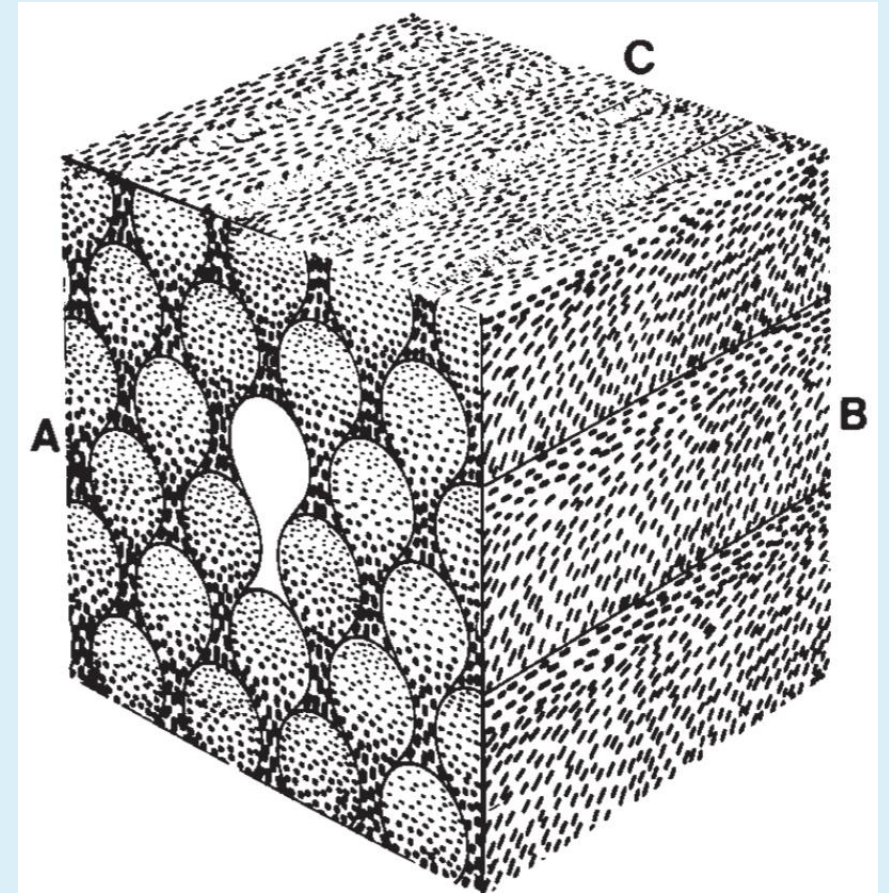
Crystallite Orientation and prism structure in a diagrammatic representation(Fig 7.13)

Cross-sectional view: characteristic keyhole arrangement (tails pointing cervically, heads occlusally)

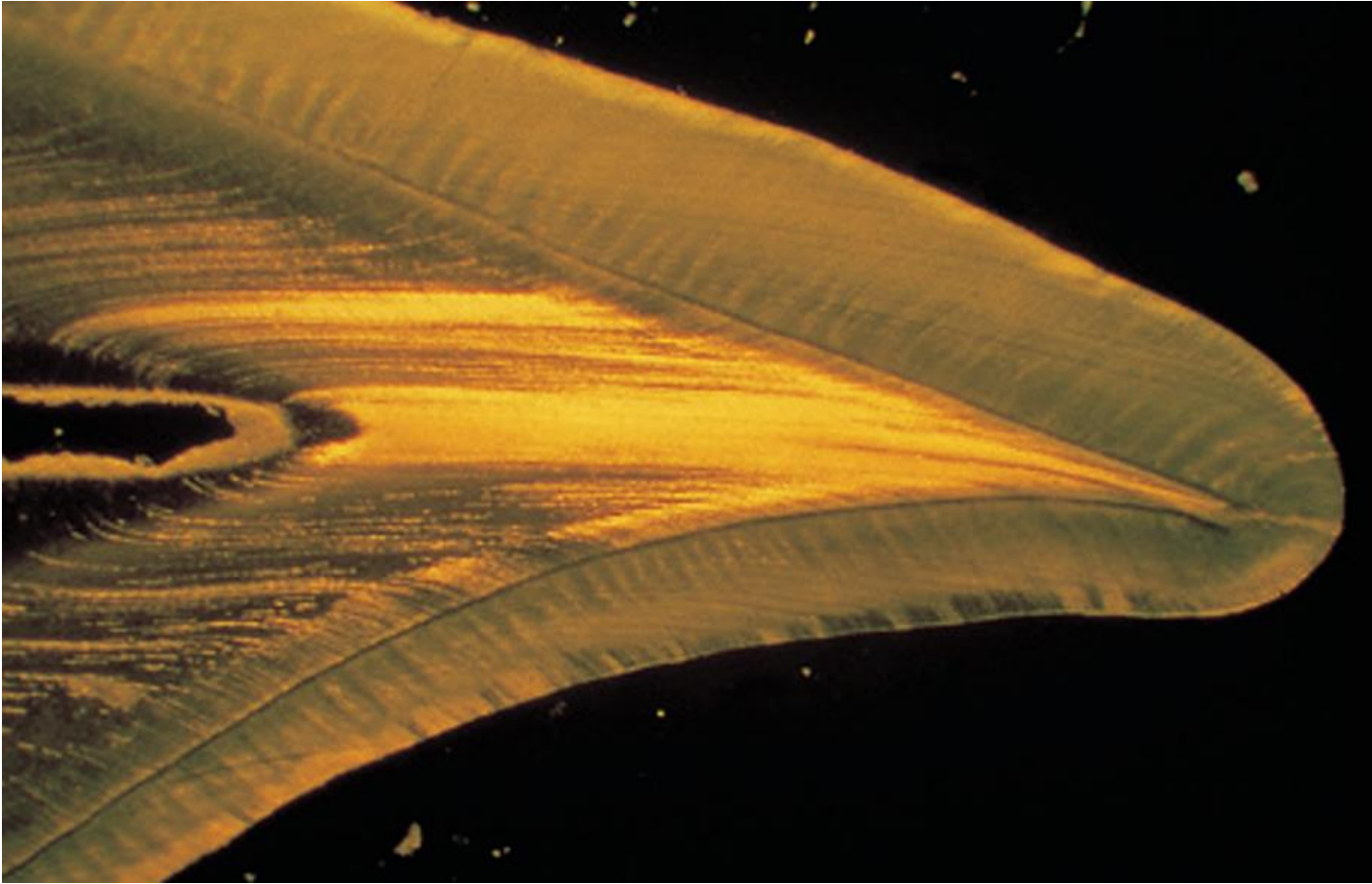
- **Prism Head** → crystallites run parallel to the long axis
- **Prism tail** → crystallites gradually diverge & become angled **65–70 degrees to the long axis**

- Within a single prism → no clear division b/w prism head & tail
- However, the crystallites in the tail of one prism show a sudden divergence from the crystallites in the head of an adjacent prism..

Interprismatic crystallites → cross prisms at an angle of approx. **60 degrees**.



A- Cross sectional view
B- Lateral surface
C- Top surface



Longitudinal section of enamel showing Hunter–Schreger bands in reflected light (7.23)

- Periodic changes in prism direction → banding pattern (50 μm wide) between 10 & 13 layers → *Hunter–Schreger bands*

- Longitudinally cut bands of prisms → *parazones*
- More transversely cut bands of prisms → *diazones*

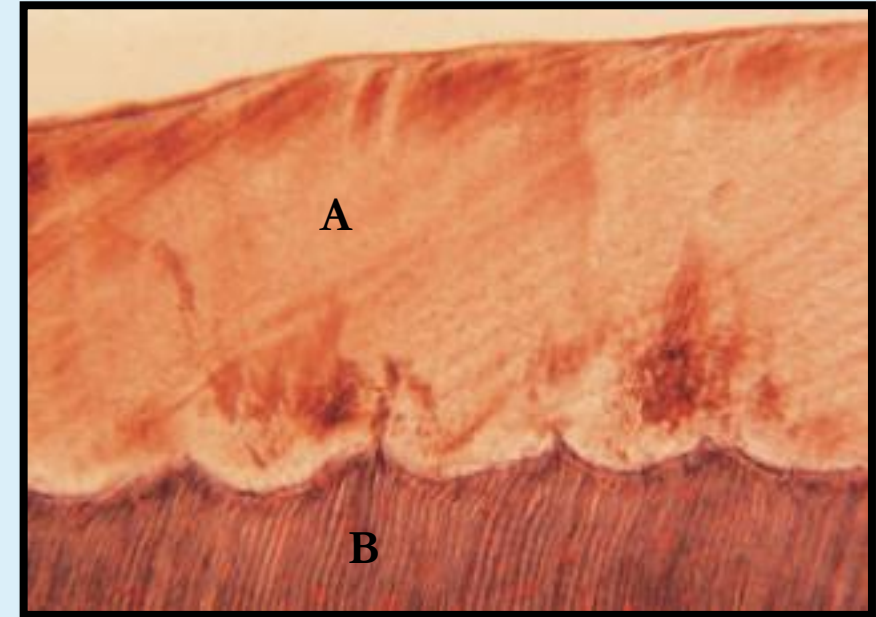
< *b/w parazones & diazones* → 40 degrees

Scalloped appearance of DEJ (Fig.52)

Scalloped 25 to 100 μm wide,

The *convexities* of the scallops are on the *enamel* surface, with the *concavities* on the *dentinal* surface

microscallop 2–5 μm in size.



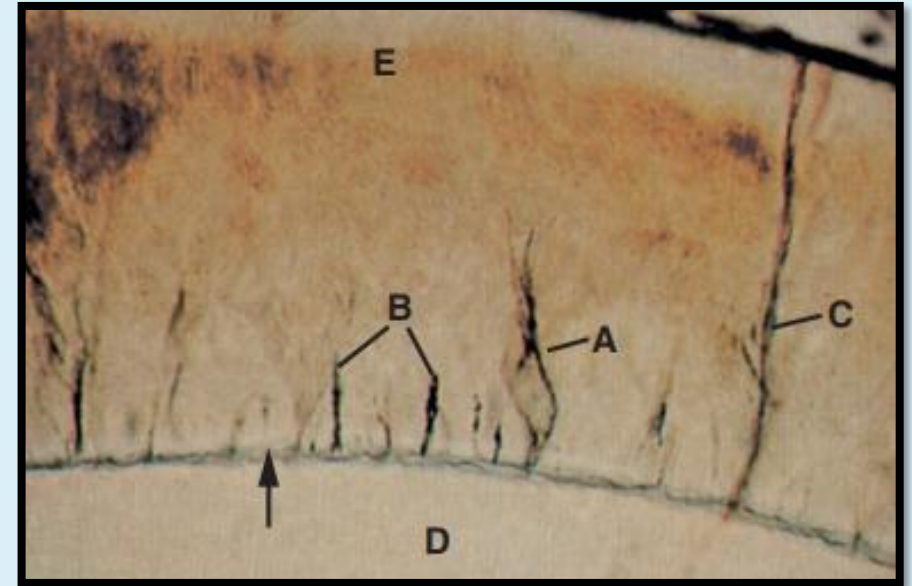
A – Enamel

B – Dentine

Transverse section showing the enamel-dentine junction (Fig.58)

Enamel Spindle

- club-shaped tubules – the enamel spindles – extend up to 25 μm into the enamel
- dentinal collagen or odontoblasts remnants



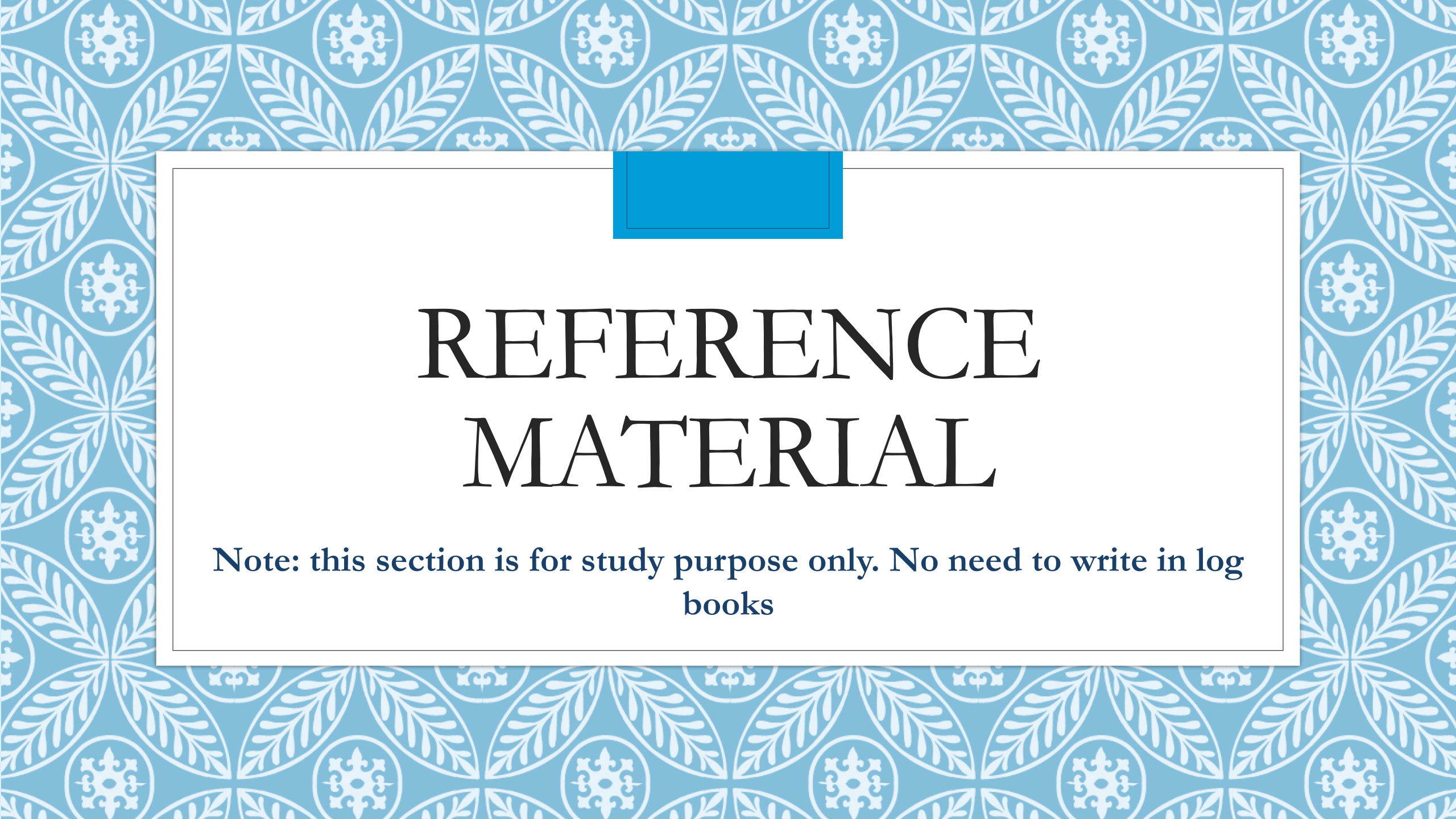
Enamel tuft

- inner third of the enamel, resemble tufts of grass
- ***Hypomineralized***
- Contain “tuftelin”
- Best visualized in ground sections

Enamel Lamella:

- sheetlike, apparent structural faults run through the entire thickness of the enamel.
- ***Hypomineralized***
- Best visualized in ground sections
- ***Incomplete enamel maturation***

A – Enamel tuft
B – Enamel Spindle
C – Enamel lamella



REFERENCE MATERIAL

Note: this section is for study purpose only. No need to write in log books

SOME FACTS ABOUT ENAMEL

Enamel *neither undergoes repair or replacement*

ORIGIN:

Ameloblasts (derive from) → “*Internal Enamel Epithelium*”

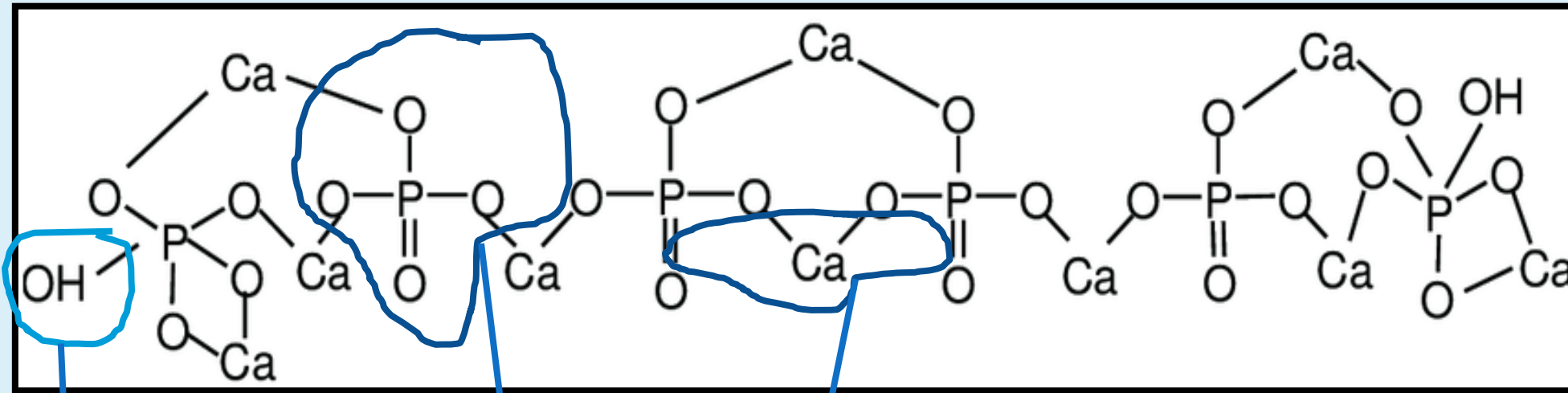
Hardness & Density ↓ :

- from surface to interior
- From cusp/incisal tip towards cervical

- Thickest over cusps/incisal edges
- Thinnest cervically
- Thickness ↑ from 1st to 3rd molar

Non-prismatic enamel -->
surface and at DEJ
More mineralized → *no prism boundaries*

Chemical formula of calcium hydroxyapatite:

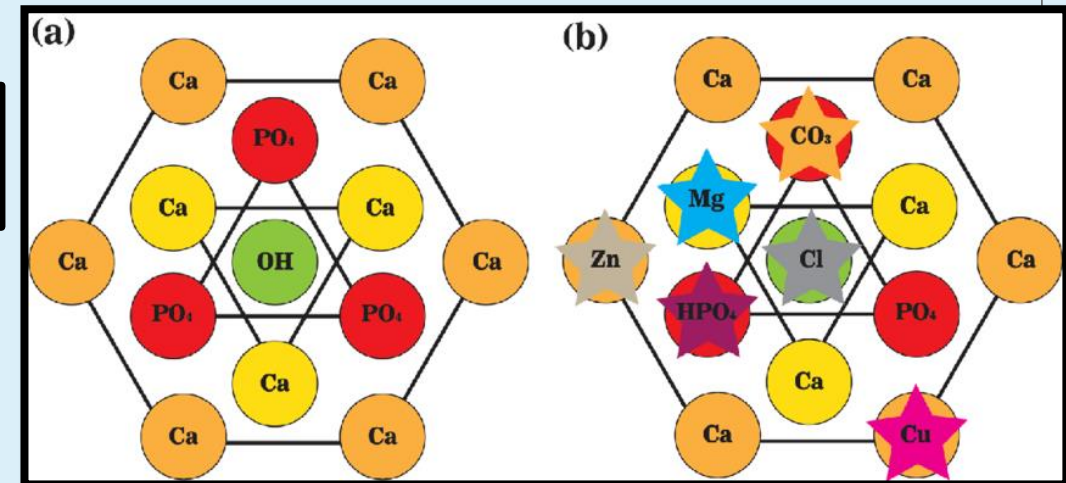


Fluoride

↑ stability & resistance

Carbonate

Magnesium



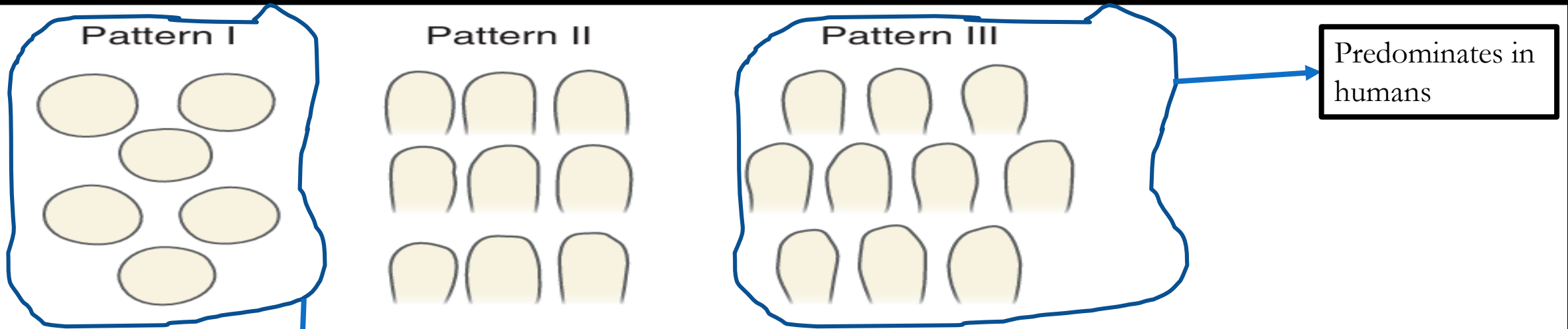
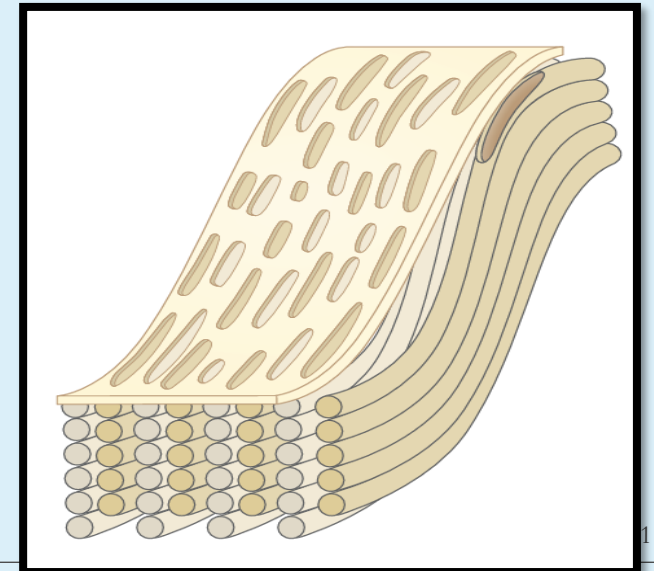
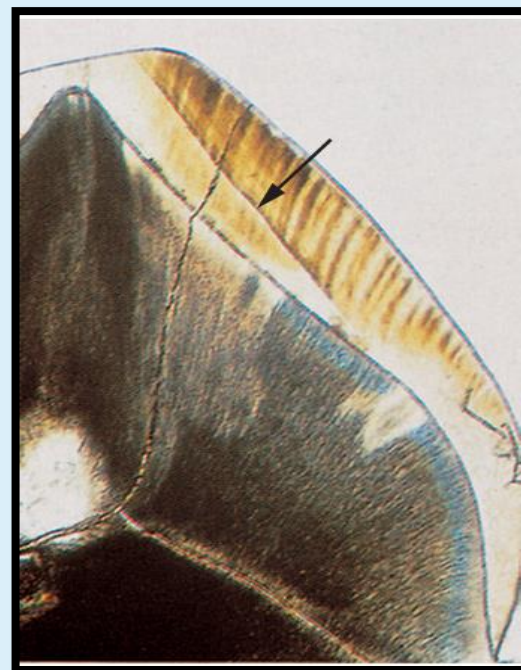
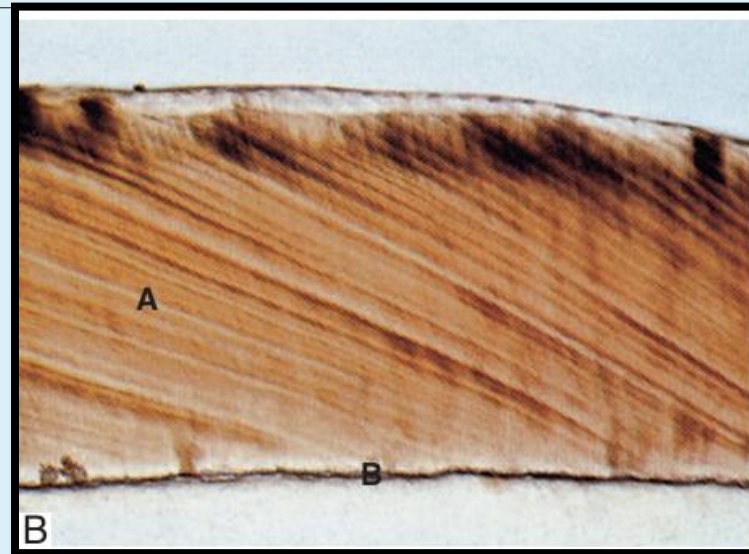


Fig. 7.10 The three prism patterns seen in human enamel. In pattern I enamel the prisms are circular. In pattern II enamel the prisms are aligned in parallel rows. In pattern III enamel the prisms are arranged in staggered rows such that the tail of a prism lies between two heads in the next row, giving a keyhole appearance.

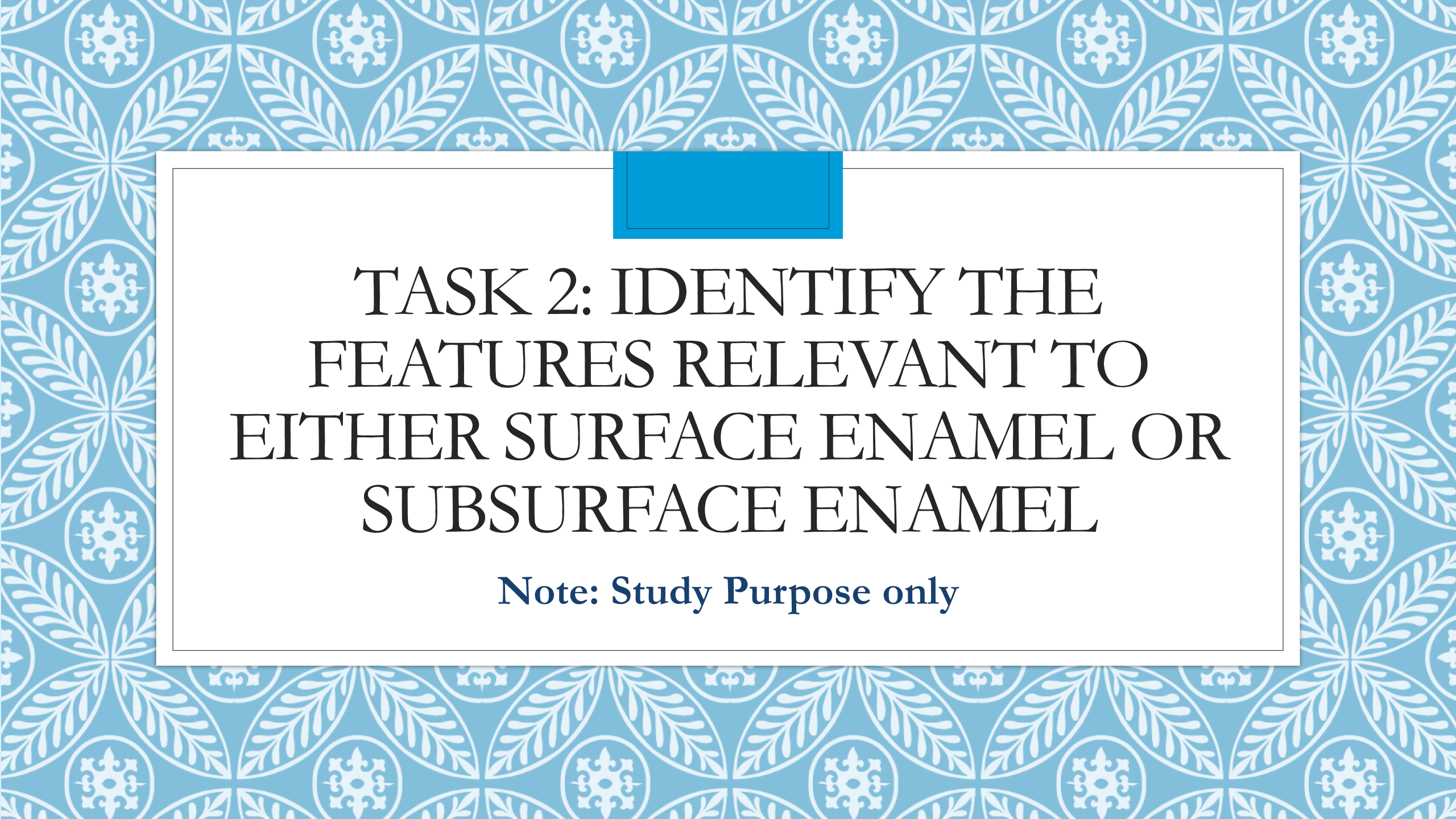
Boundary complete & circular (found near DEJ) ^
surface b/c deposition is slow

*In enamel fractured or sectioned parallel to long axis of the tooth → prisms travel in a **sinusoidal line** from the DEJ to the surface*



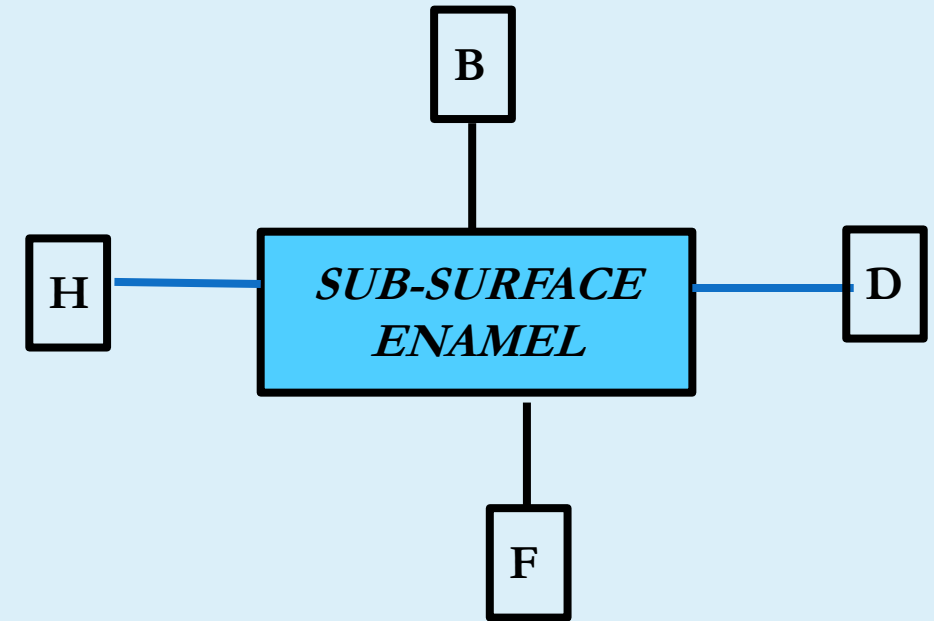
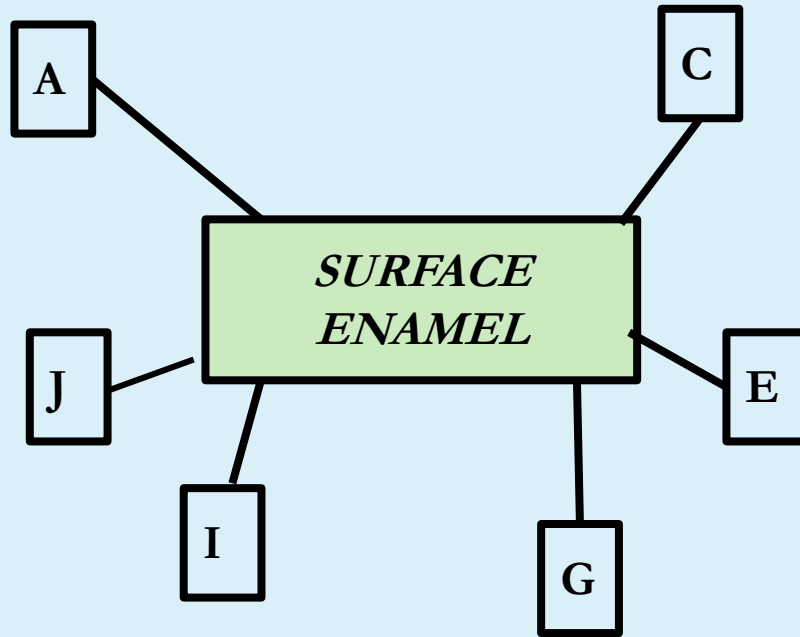




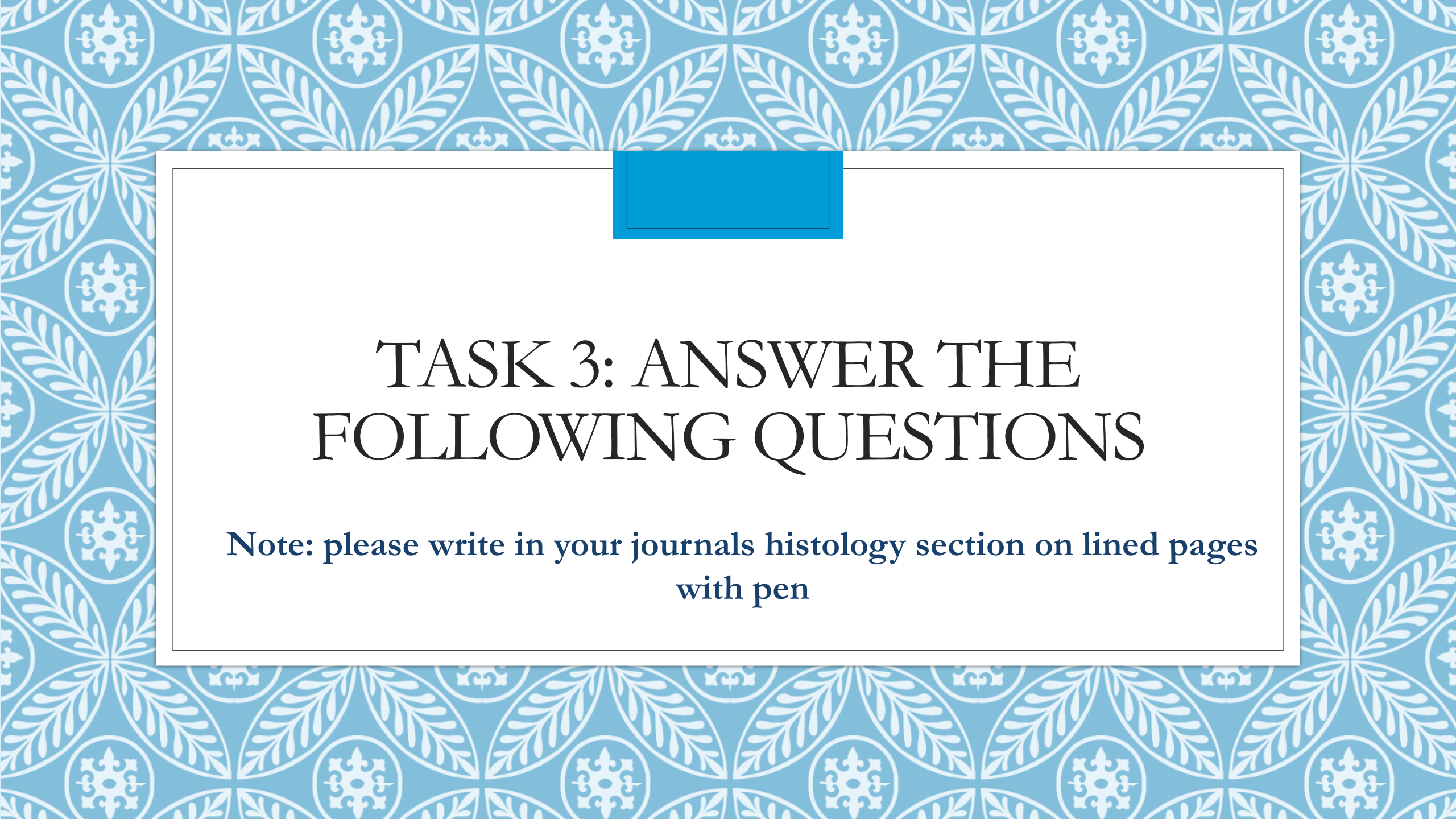


TASK 2: IDENTIFY THE FEATURES RELEVANT TO EITHER SURFACE ENAMEL OR SUBSURFACE ENAMEL

Note: Study Purpose only



A	Harder and denser	F	Prismatic structure
B	More porous	G	More radio-opaque
C	Higher fluoride content	H	Less resistant to caries
D	More soluble	I	Directly exposed to oral environment
E	Mostly aprismatic	J	Less carbonate content



TASK 3: ANSWER THE FOLLOWING QUESTIONS

**Note: please write in your journals histology section on lined pages
with pen**

What is the chemical composition of enamel?

The principal mineral component of enamel is calcium hydroxyapatite $[\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2]$ in crystalline form, making up for 95–96% by weight of inorganic. Water constitutes about 2% by weight of enamel and only 1–2% by weight of organic matrix.

What are incremental lines and its types?

Box 7.1 Lines in enamel

Cross striations	Enamel stria	Neonatal line
Short period incremental lines	Long period incremental lines	Reflects metabolic changes at birth
Transverse to prisms	Oblique to prisms	Oblique to prisms
Throughout enamel	Throughout enamel	A single line
4 μm apart	25–40 μm apart	About 20–40 μm thick
Represent daily increments	Represent weekly increments	Represents a few days after birth
Prism varies in width	Reach surface at perikymata	Prisms vary in width and direction

What are the age-related changes that appear in enamel?

Enamel wears away slowly with age, the amount lost being dependent on diet and masticatory habits. It seems to darken in colour, due to the reduced translucency of the tooth as secondary dentine forms and enamel thins.

What is aprismatic/prismless enamel?

Non-prismatic enamel occurs as a result of the absence of Tomes processes from the ameloblasts in the first and final stages of enamel deposition. Prismless enamel is highly mineralized and present at the DEJ and the surface enamel.

**THANK
YOU!**

The image features the words "THANK YOU!" in a bold, yellow, sans-serif font with a thick black outline. The text is arranged in two lines: "THANK" on top and "YOU!" on the bottom. The exclamation mark is red with a black outline. Radiating from the text are numerous short, black, dashed lines of varying lengths, creating a starburst or explosion effect. The entire graphic is centered on a plain white background.